

Technology Stack

Date	15 March 2026
Team ID	-
Project Name	Flood Prediction Using Machine Learning
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Builds a responsive and interactive user interface.
2	Backend / Server-Side	Python Flask	Handles ML prediction requests, API logic, and data processing.
3	Database / Data Storage	CSV Dataset (Kaggle)	Stores historical flood data used for training and prediction.
4	Cloud / Hosting / Deployment	Visual Studio Code (Development Environment)	Used to develop, test, and run the application locally.
5	Version Control & CI/CD	Git & GitHub	Tracks code changes and enables team collaboration.

Dataset Source: Kaggle

Machine Learning Libraries: Scikit-learn, Pandas, NumPy, Category_encoders, pickle, Flask